CSA 4302 - INTERNET PROGRAMMING

LAB EXPERIMENT 02:

AIM : To design and develop a responsive Online Hotel Room Booking Website that enables users to search for hotels, check room availability, book rooms for specific dates, manage their profiles, and view booking history. The system allows users to modify or cancel bookings up to three days before the arrival date. It also provides an administrator panel to manage room types, prices, availability, booking details, and website content, thereby improving the efficiency and convenience of hotel booking operations.

CODE:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Cornerstone Library</title>
  <style>
    :root {
      color-scheme: light;
      --ink: #24312d;
      --muted: #68756f;
      --paper: #f7f8f4;
      --card: #fff;
      --green: #27634f;
      --green-dark: #194b3a;
      --line: #e3e9e3;
      --accent: #eaf2ed;
      --danger: #a43c34;
      font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont,
"Segoe UI", sans-serif;
    }
    * { box-sizing: border-box; }
    body { margin: 0; background: var(--paper); color: var(--ink); }
    button, input, select, textarea { font: inherit; }
    button { cursor: pointer; }
    header { background: var(--card); border-bottom: 1px solid var(--line); }
    .header-inner { max-width: 1100px; margin: auto; padding: 16px 24px; display:
flex; align-items: center; justify-content: space-between; gap: 18px; }
    .brand { font-size: 1.15rem; font-weight: 750; color: var(--green-dark); letter-
spacing: -.03em; }
    .brand span { color: #9a7952; }
    nav { display: flex; gap: 7px; flex-wrap: wrap; }
```

```
    nav button, .link-button { background: transparent; border: 0; color:
var(--muted); padding: 9px 11px; border-radius: 8px; }
    nav button:hover, nav button.active { background: var(--accent); color: var(--green-dark); }
    .header-actions { display: flex; align-items: center; gap: 10px; }
    .user-label { color: var(--muted); font-size: .9rem; }
    .container { max-width: 1100px; padding: 36px 24px 60px; margin: auto; }
    .hero { display: grid; grid-template-columns: 1.4fr .8fr; gap: 28px; align-items:
center; background: #eaf1e9; border-radius: 18px; padding: 34px; margin-bottom: 30px;
}
    .eyebrow { color: var(--green); text-transform: uppercase; letter-spacing: .12em;
font-size: .75rem; font-weight: 700; }
    h1 { font-size: clamp(2rem, 5vw, 3.2rem); line-height: 1.05; letter-spacing: -
.05em; margin: 10px 0 14px; }
    h2 { letter-spacing: -.03em; margin: 0 0 8px; }
    h3 { margin: 0 0 8px; }
    p { color: var(--muted); line-height: 1.6; }
    .hero p { max-width: 530px; margin: 0 0 20px; }
    .hero-art { min-height: 185px; border-radius: 14px; background:
linear-gradient(145deg, #cbdccd, #e8d9bd); display: grid; place-items: center; font-
size: 5rem; }
    .button { border: 1px solid var(--green); border-radius: 8px; padding: 9px 13px;
background: var(--green); color: white; font-weight: 650; }
    .button:hover { background: var(--green-dark); }
    .button.secondary { color: var(--green-dark); background: white; }
    .button.secondary:hover { background: var(--accent); }
    .button.danger { background: white; color: var(--danger); border-color: #e5b9b5; }
    .button.small { padding: 6px 9px; font-size: .86rem; }
    .section-heading { display: flex; justify-content: space-between; align-items:
end; gap: 14px; margin: 28px 0 15px; }
    .section-heading p { margin: 4px 0 0; }
    .grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 16px; }
    .card { background: var(--card); border: 1px solid var(--line); border-radius:
12px; padding: 18px; }
    .book-card { display: flex; flex-direction: column; min-height: 195px; }
    .book-card .meta { color: var(--muted); font-size: .9rem; }
    .book-card .availability { margin-top: auto; padding-top: 14px; display: flex;
justify-content: space-between; align-items: center; gap: 12px; }
    .badge { border-radius: 99px; background: var(--accent); color: var(--green-dark);
padding: 4px 9px; font-size: .78rem; font-weight: 650; }
    .badge.unavailable { color: var(--danger); background: #f9e9e7; }
    .toolbar { display: flex; gap: 10px; margin-bottom: 16px; }
    input, select, textarea { width: 100%; padding: 10px 11px; border-radius: 8px;
border: 1px solid #d7dfd8; background: white; color: var(--ink); }
    input:focus, select:focus, textarea:focus { outline: 2px solid #bad5c6; border-
color: var(--green); }
    label { display: grid; gap: 6px; color: #43514b; font-size: .9rem; font-weight:
600; }
    form { display: grid; gap: 12px; }
```

```
    .form-grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 12px; }
    .stack { display: grid; gap: 16px; }
    .table-wrap { overflow-x: auto; }
    table { border-collapse: collapse; width: 100%; text-align: left; font-size:
.91rem; }
    th, td { padding: 11px 10px; border-bottom: 1px solid var(--line); vertical-align:
middle; }
    th { color: var(--muted); font-weight: 650; font-size: .82rem; }
    .empty { padding: 26px; text-align: center; color: var(--muted); background:
white; border: 1px dashed #cdd8cf; border-radius: 10px; }
    .notice { padding: 12px 14px; border-radius: 8px; background: var(--accent);
color: var(--green-dark); margin-bottom: 14px; }
    .notice.error { background: #f9e9e7; color: var(--danger); }
    .fine { font-weight: 700; color: var(--danger); }
    .hidden { display: none !important; }
    .modal { position: fixed; inset: 0; z-index: 3; background: #15231dcc; display:
grid; place-items: center; padding: 18px; }
    .modal-card { width: min(440px, 100%); background: white; border-radius: 14px;
padding: 24px; box-shadow: 0 16px 60px #0003; }
    .modal-head { display: flex; justify-content: space-between; align-items: center;
margin-bottom: 16px; }
    .close { border: 0; background: transparent; font-size: 1.4rem; color:
var(--muted); }
    footer { border-top: 1px solid var(--line); padding: 23px; text-align: center;
color: var(--muted); font-size: .88rem; }
    @media (max-width: 760px) {
      .header-inner { align-items: flex-start; flex-wrap: wrap; }
      .header-actions { margin-left: auto; }
      nav { order: 3; width: 100%; }
      .container { padding: 25px 16px 45px; }
      .hero { grid-template-columns: 1fr; padding: 25px; }
      .hero-art { min-height: 130px; }
      .grid { grid-template-columns: repeat(2, 1fr); }
    }
    @media (max-width: 520px) {
      .grid, .form-grid { grid-template-columns: 1fr; }
      .toolbar { flex-direction: column; }
      .header-inner { padding: 13px 16px; }
      nav { overflow-x: auto; flex-wrap: nowrap; }
      nav button { white-space: nowrap; }
      .user-label { display: none; }
    }
  </style>
</head>
<body>
  <header>
    <div class="header-inner">
      <div class="brand">Cornerstone <span>Library</span></div>
```

```
        <nav aria-label="Main navigation">
          <button class="active" data-page="home">Home</button>
          <button data-page="catalog">Book catalog</button>
          <button data-page="materials">Study materials</button>
          <button data-page="dashboard" class="hidden" id="dashboard-nav">My
account</button>
          <button data-page="admin" class="hidden" id="admin-nav">Librarian</button>
          <button data-page="contact">Contact</button>
        </nav>
        <div class="header-actions">
          <span class="user-label" id="user-label"></span>
          <button class="button secondary small" id="login-open">Log in</button>
          <button class="button small hidden" id="logout">Log out</button>
        </div>
      </div>
    </header>
    <main class="container">
      <div id="message" class="notice hidden" role="status"></div>
      <section id="page-home" class="page">
        <div class="hero">
          <div>
            <div class="eyebrow">Your campus library</div>
            <h1>Find your next<br>great read.</h1>
            <p>Search the library catalog, explore study materials, and keep track of
the books you borrow—all in one place.</p>
            <button class="button" data-go="catalog">Explore the catalog</button>
          </div>
          <div class="hero-art" aria-hidden="true">📚</div>
        </div>
        <div class="section-heading">
          <div><h2>Popular books</h2><p>A few picks from our library shelves.</p></div>
          <button class="link-button" data-go="catalog">View all books →</button>
        </div>
        <div id="featured-books" class="grid"></div>
      </section>
      <section id="page-catalog" class="page hidden">
        <div class="section-heading"><div><div class="eyebrow">Browse</div><h2>Book
catalog</h2><p>Search by title, author, or subject.</p></div></div>
        <div class="toolbar"><input id="book-search" type="search" placeholder="Search
books…" aria-label="Search books"></div>
        <div id="book-results" class="grid"></div>
      </section>
      <section id="page-materials" class="page hidden">
        <div class="section-heading"><div><div class="eyebrow">For your
studies</div><h2>Study materials</h2><p>Find resources by academic year and
semester.</p></div></div>
        <div class="toolbar">
```

```
        <select id="material-year" aria-label="Filter by year"><option value="">All years</option></select>
        <select id="material-semester" aria-label="Filter by semester"><option value="">All semesters</option></select>
      </div>
      <div id="material-results" class="grid"></div>
    </section>
    <section id="page-dashboard" class="page hidden">
      <div class="section-heading"><div><div class="eyebrow">Your account</div><h2>My library</h2><p id="account-greeting"></p></div></div>
      <div class="card" style="margin-bottom:16px"><h3>My profile</h3><div id="profile-details"></div></div>
      <div class="card"><h3>Borrow history</h3><p>Current overdue fines are calculated at $1 per day.</p><div id="history-table" class="table-wrap"></div></div>
    </section>
    <section id="page-admin" class="page hidden">
      <div class="section-heading"><div><div class="eyebrow">Library management</div><h2>Librarian dashboard</h2><p>Manage books, loans, and study materials.</p></div></div>
      <div class="stack">
        <div class="card"><h3>Add a book</h3>
          <form id="book-form">
            <input type="hidden" name="bookId">
            <div class="form-grid"><label>Book title<input name="title" required></label><label>Author<input name="author" required></label><label>Subject<input name="subject" required></label><label>Copies<input name="copies" type="number" min="1" value="1" required></label></div>
            <div><button class="button" type="submit">Add book</button> <button class="button secondary hidden" id="cancel-book-edit" type="button">Cancel edit</button></div>
          </form>
        </div>
        <div class="card"><h3>Inventory</h3><div id="inventory-table" class="table-wrap"></div></div>
        <div class="card"><h3>Issue a book</h3>
          <form id="issue-form">
            <div class="form-grid"><label>Student account<select name="user" id="student-select" required></select></label><label>Available book<select name="book" id="issue-book-select" required></select></label><label>Expected return date<input name="due" type="date" required></label></div>
            <button class="button" type="submit">Issue book</button>
          </form>
        </div>
        <div class="card"><h3>Issued books &amp;amp; returns</h3><div id="loans-table" class="table-wrap"></div></div>
        <div class="card"><h3>Add study material</h3>
          <form id="material-form">
```

```
            <input type="hidden" name="materialId">
            <div class="form-grid"><label>Material title<input name="title"
required></label><label>Year<input name="year" type="number" min="1" max="8"
placeholder="e.g. 1" required></label><label>Semester<input name="semester"
type="number" min="1" max="12" placeholder="e.g. 1" required></label><label>File<input
name="file" type="file" accept=".pdf,.doc,.docx,.ppt,.pptx,.txt"></label></div>
            <div><button class="button" type="submit">Add material</button> <button
class="button secondary hidden" id="cancel-material-edit" type="button">Cancel
edit</button></div>
          </form>
          <div id="admin-materials" class="table-wrap" style="margin-top:18px"></div>
        </div>
      </div>
    </section>
    <section id="page-contact" class="page hidden">
      <div class="section-heading"><div><div class="eyebrow">Get in
touch</div><h2>Contact the library</h2><p>Send the librarian a
message.</p></div></div>
      <div class="card" style="max-width:620px">
        <form id="contact-form">
          <label>Your name<input name="name" required></label><label>Email<input
type="email" name="email" required></label><label>Message<textarea name="message"
rows="5" required></textarea></label>
          <button class="button" type="submit">Send message</button>
        </form>
      </div>
    </section>
  </main>
  <footer>Cornerstone Library · Open for your next chapter</footer>
  <div id="auth-modal" class="modal hidden" role="dialog" aria-modal="true" aria-
labelledby="auth-title">
    <div class="modal-card">
      <div class="modal-head"><h2 id="auth-title">Log in</h2><button class="close"
id="modal-close" aria-label="Close">×</button></div>
      <div class="toolbar"><button class="button small" id="show-login">Log
in</button><button class="button secondary small" id="show-register">Create
account</button></div>
      <form id="login-form"><label>Email<input name="email" type="email" required
autocomplete="username"></label><label>Password<input name="password" type="password"
required autocomplete="current-password"></label><button class="button"
type="submit">Log in</button></form>
      <form id="register-form" class="hidden"><label>Name<input name="name" required
autocomplete="name"></label><label>Email<input name="email" type="email" required
autocomplete="email"></label><label>Password<input name="password" type="password"
minlength="4" required autocomplete="new-password"></label><button class="button"
type="submit">Create account</button></form>
      <p style="font-size:.82rem;margin-bottom:0">Demo librarian:
librarian@library.test / library123</p>
```

```
    </div>
  </div>
  <script>
    const KEY = "cornerstone-library-v1";
    const initial = {
      users: [
        { name: "Library Admin", email: "librarian@library.test", password:
"library123", role: "admin" }
      ],
      books: [
        { id: 1, title: "The Great Gatsby", author: "F. Scott Fitzgerald", subject:
"Literature", copies: 3 },
        { id: 2, title: "A Brief History of Time", author: "Stephen Hawking", subject:
"Science", copies: 2 },
        { id: 3, title: "Introduction to Algorithms", author: "Thomas H. Cormen",
subject: "Computer Science", copies: 4 },
        { id: 4, title: "Pride and Prejudice", author: "Jane Austen", subject:
"Literature", copies: 2 },
        { id: 5, title: "The Selfish Gene", author: "Richard Dawkins", subject:
"Biology", copies: 1 },
        { id: 6, title: "The Art of Statistics", author: "David Spiegelhalter",
subject: "Mathematics", copies: 2 }
      ],
      loans: [],
      materials: [
        { id: 1, title: "Introduction to Biology – Notes", year: "1", semester: "1",
filename: "biology-notes.txt", data: "Welcome to the library study materials. Replace
this sample with your course notes." },
        { id: 2, title: "Calculus – Revision Guide", year: "1", semester: "2",
filename: "calculus-guide.txt", data: "Sample revision guide. Librarians can add real
course files from the dashboard." },
        { id: 3, title: "Database Systems – Lecture Notes", year: "2", semester: "1",
filename: "database-notes.txt", data: "Sample lecture notes. Upload a PDF or document
in the librarian dashboard." }
      ]
    };
    let db;
    try {
      db = JSON.parse(localStorage.getItem(KEY)) || initial;
      if (!Array.isArray(db.users) || !Array.isArray(db.books) ||
!Array.isArray(db.loans) || !Array.isArray(db.materials)) throw new Error("Invalid
saved library data");
    } catch (error) {
      console.error("Could not load saved library data:", error);
      db = JSON.parse(JSON.stringify(initial));
    }
    let currentUser = null;
    const $ = (selector) => document.querySelector(selector);
```

```
    const escapeHTML = (value) => String(value).replace(/[&<>"']/g, c => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", '"': "&quot;", "'": "&#39;" })[c]);
    function save() { localStorage.setItem(KEY, JSON.stringify(db)); }
    function notice(text, error = false) {
      const el = $("#message"); el.textContent = text; el.classList.remove("hidden", "error");
      if (error) el.classList.add("error");
      window.scrollTo({ top: 0, behavior: "smooth" });
      setTimeout(() => el.classList.add("hidden"), 4500);
    }
    function available(book) {
      return book.copies - db.loans.filter(loan => loan.bookId === book.id && !loan.returned).length;
    }
    function fine(loan) {
      if (loan.returned) return 0;
      const days = Math.floor((Date.now() - new Date(loan.due + "T23:59:59").getTime()) / 86400000);
      return Math.max(0, days) * 1;
    }
    function go(page) {
      document.querySelectorAll(".page").forEach(section => section.classList.toggle("hidden", section.id !== `page-${page}`));
      document.querySelectorAll("nav [data-page]").forEach(button => button.classList.toggle("active", button.dataset.page === page));
      if (page === "catalog") renderBooks();
      if (page === "materials") renderMaterials();
      if (page === "dashboard") renderDashboard();
      if (page === "admin") renderAdmin();
      window.scrollTo(0, 0);
    }
    function bookCard(book) {
      const count = available(book);
      return `<article class="card book-card"><h3>${escapeHTML(book.title)}</h3><div class="meta">by ${escapeHTML(book.author)}</div><div class="meta">${escapeHTML(book.subject)}</div><div class="availability"><span class="badge ${count < 1 ? "unavailable" : ""}">${count} available</span><button class="button secondary small details-button" data-id="${book.id}">Details</button></div></article>`;
    }
    function renderBooks() {
      const query = ($("#book-search").value || "").trim().toLowerCase();
      const books = db.books.filter(book => `${book.title} ${book.author} ${book.subject}`.toLowerCase().includes(query));
      $("#book-results").innerHTML = books.length ? books.map(bookCard).join("") : '<div class="empty">No books found. Try another search.</div>';
      document.querySelectorAll(".details-button").forEach(button => button.addEventListener("click", () => {
```

```
        const book = db.books.find(item => item.id === Number(button.dataset.id));
        if (book) notice(`${book.title} by ${book.author} · ${book.subject} · ${available(book)} copies available.`);
      }));
    }
    function renderMaterials() {
      const years = [...new Set(db.materials.map(item => item.year))].sort((a,b) => Number(a)-Number(b));
      const semesters = [...new Set(db.materials.map(item => item.semester))].sort((a,b) => Number(a)-Number(b));
      const year = $("#material-year").value, semester = $("#material-semester").value;
      $("#material-year").innerHTML = '<option value="">All years</option>' + years.map(value => `<option value="${escapeHTML(value)}">${escapeHTML(value)}</option>`).join("");
      $("#material-semester").innerHTML = '<option value="">All semesters</option>' + semesters.map(value => `<option value="${escapeHTML(value)}">${escapeHTML(value)}</option>`).join("");
      $("#material-year").value = year; $("#material-semester").value = semester;
      const filtered = db.materials.filter(item => (!year || item.year === year) && (!semester || item.semester === semester));
      $("#material-results").innerHTML = filtered.length ? filtered.map(item => `<article class="card"><div class="eyebrow">Year ${escapeHTML(item.year)} · Semester ${escapeHTML(item.semester)}</div><h3 style="margin-top:10px">${escapeHTML(item.title)}</h3><a class="button secondary small" style="display:inline-block;text-decoration:none;margin-top:10px" href="${item.data || ""}" download="${escapeHTML(item.filename)}">Download material</a></article>`).join("") : '<div class="empty">No materials match these filters.</div>';
    }
    function renderDashboard() {
      if (!currentUser || currentUser.role !== "user") return go("home");
      $("#account-greeting").textContent = `Welcome back, ${currentUser.name}.`;
      $("#profile-details").innerHTML = `<p><strong>Name:</strong> ${escapeHTML(currentUser.name)}<br><strong>Email:</strong> ${escapeHTML(currentUser.email)}</p>`;
      const loans = db.loans.filter(loan => loan.email === currentUser.email).sort((a,b) => b.issue.localeCompare(a.issue));
      $("#history-table").innerHTML = loans.length ? `<table><thead><tr><th>Book</th><th>Issue date</th><th>Due date</th><th>Status</th><th>Fine</th></tr></thead><tbody>${loans.map(loan => `<tr><td>${escapeHTML(db.books.find(book => book.id === loan.bookId)?.title || "Book removed")}</td><td>${escapeHTML(loan.issue)}</td><td>${escapeHTML(loan.due)}</td><td>${loan.returned ? "Returned" : (fine(loan) ? "Overdue" : "Issued")}</td><td class="${fine(loan) ? "fine" : ""}">$${fine(loan)}</td></tr>`).join("")}</tbody></table>` : '<div class="empty">No borrowed books yet.</div>';
    }
```

```
    function renderAdmin() {
      if (!currentUser || currentUser.role !== "admin") return go("home");
      $("#inventory-table").innerHTML =
`<table><thead><tr><th>Book</th><th>Author</th><th>Copies</th><th>Available</th><th>Ac
tion</th></tr></thead><tbody>${db.books.map(book =>
`<tr><td>${escapeHTML(book.title)}</td><td>${escapeHTML(book.author)}</td><td>${book.c
opies}</td><td>${available(book)}</td><td><button class="button secondary small edit-
book" data-id="${book.id}">Edit</button> <button class="button danger small delete-
book" data-id="${book.id}">Delete</button></td></tr>`).join("") || '<tr><td
colspan="5">No books in inventory.</td></tr>'}</tbody></table>`;
      document.querySelectorAll(".edit-book").forEach(button =>
button.addEventListener("click", () => {
        const book = db.books.find(item => item.id === Number(button.dataset.id));
        if (!book) return;
        const form = $("#book-form");
        form.elements.bookId.value = book.id;
        form.elements.title.value = book.title;
        form.elements.author.value = book.author;
        form.elements.subject.value = book.subject;
        form.elements.copies.min = book.copies - available(book);
        form.elements.copies.value = book.copies;
        form.querySelector('[type="submit"]').textContent = "Update book";
        $("#cancel-book-edit").classList.remove("hidden");
        form.scrollIntoView({ behavior: "smooth", block: "center" });
      }));
      document.querySelectorAll(".delete-book").forEach(button =>
button.addEventListener("click", () => {
        const id = Number(button.dataset.id);
        if (db.loans.some(loan => loan.bookId === id && !loan.returned)) return
notice("Return all copies before deleting this book.", true);
        db.books = db.books.filter(book => book.id !== id); save(); renderAdmin();
renderBooks();
      }));
      const students = db.users.filter(user => user.role === "user");
      $("#student-select").innerHTML = students.map(user => `<option
value="${escapeHTML(user.email)}">${escapeHTML(user.name)}
(${escapeHTML(user.email)})</option>`).join("");
      $("#issue-book-select").innerHTML = db.books.filter(book => available(book) >
0).map(book => `<option value="${book.id}">${escapeHTML(book.title)}
(${available(book)} available)</option>`).join("");
      $("#issue-form button[type=submit]").disabled = !students.length ||
!db.books.some(book => available(book) > 0);
      const loans = [...db.loans].sort((a,b) => b.issue.localeCompare(a.issue));
      $("#loans-table").innerHTML = loans.length ?
`<table><thead><tr><th>Student</th><th>Book</th><th>Issued</th><th>Due</th><th>Fine</t
h><th>Status</th></tr></thead><tbody>${loans.map(loan =>
`<tr><td>${escapeHTML(loan.email)}</td><td>${escapeHTML(db.books.find(book => book.id
=== loan.bookId)?.title || "Book
```

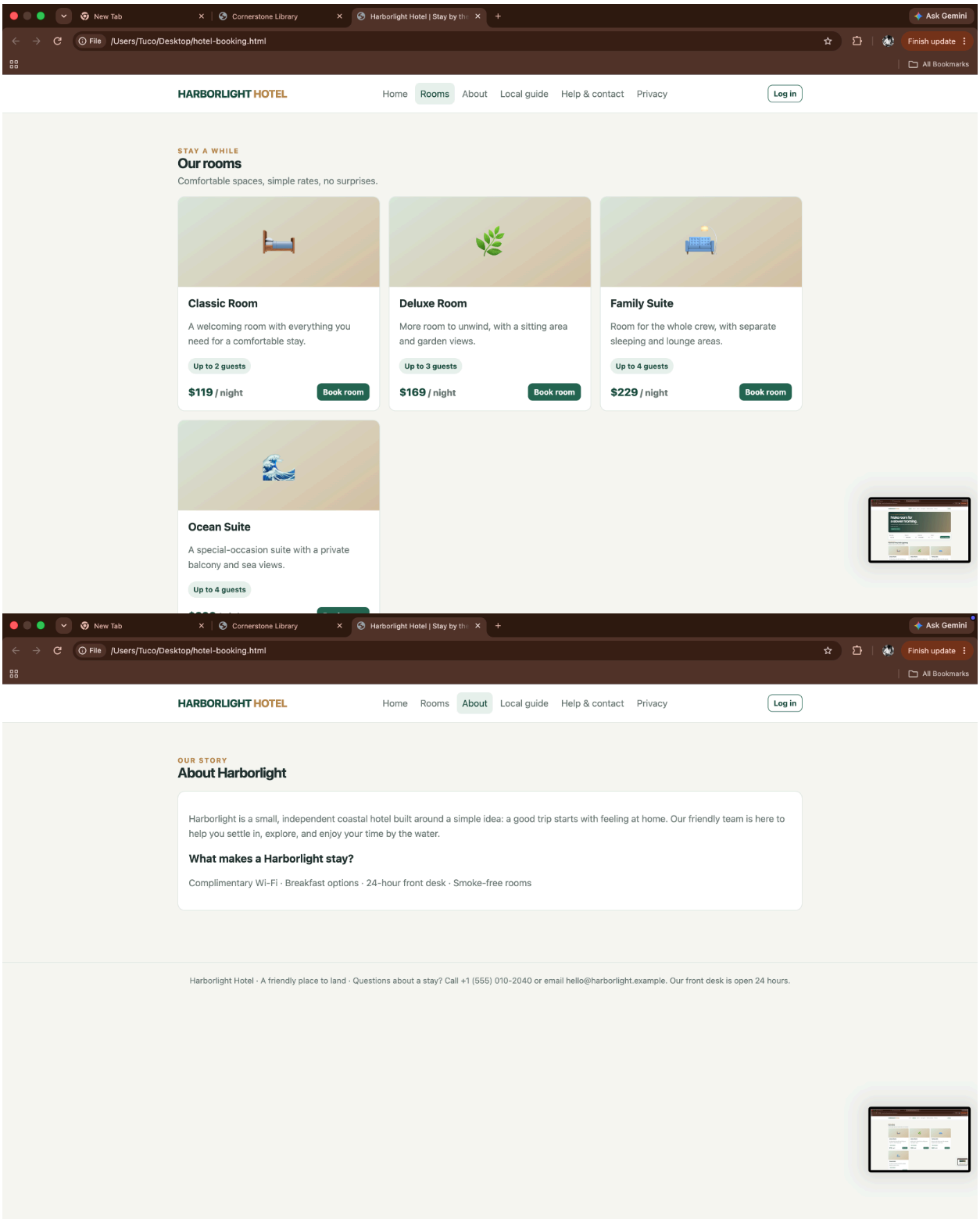
```
removed")}</td><td>${escapeHTML(loan.issue)}</td><td>${escapeHTML(loan.due)}</td><td
class="${fine(loan) ? "fine" : ""}">$${fine(loan)}</td><td>${loan.returned ?
"Returned" : `<button class="button secondary small return-book"
data-id="${loan.id}">Mark returned</button>`}</td></tr>`).join("")}</tbody></table>` :
'<div class="empty">No books have been issued.</div>';
      document.querySelectorAll(".return-book").forEach(button =>
button.addEventListener("click", () => {
        const loan = db.loans.find(item => item.id === Number(button.dataset.id));
        if (loan) { loan.returned = true; loan.returnedOn = new
Date().toISOString().slice(0,10); save(); renderAdmin(); }
      }));
      $("#admin-materials").innerHTML = db.materials.length ?
`<table><thead><tr><th>Material</th><th>Year</th><th>Semester</th><th>Actions</th></tr
></thead><tbody>${db.materials.map(item =>
`<tr><td>${escapeHTML(item.title)}</td><td>${escapeHTML(item.year)}</td><td>${escapeHT
ML(item.semester)}</td><td><button class="button secondary small edit-material" data-
id="${item.id}">Edit</button> <button class="button danger small delete-material"
data-id="${item.id}">Delete</button></td></tr>`).join("")}</tbody></table>` : '<div
class="empty">No study materials added.</div>';
      document.querySelectorAll(".edit-material").forEach(button =>
button.addEventListener("click", () => {
        const material = db.materials.find(item => item.id ===
Number(button.dataset.id));
        if (!material) return;
        const form = $("#material-form");
        form.elements.materialId.value = material.id;
        form.elements.title.value = material.title;
        form.elements.year.value = material.year;
        form.elements.semester.value = material.semester;
        form.querySelector('[type="submit"]').textContent = "Update material";
        $("#cancel-material-edit").classList.remove("hidden");
        form.scrollIntoView({ behavior: "smooth", block: "center" });
      }));
      document.querySelectorAll(".delete-material").forEach(button =>
button.addEventListener("click", () => {
        db.materials = db.materials.filter(item => item.id !==
Number(button.dataset.id));
        save(); renderAdmin(); renderMaterials();
      }));
    }
    function updateAuth() {
      const loggedIn = !!currentUser;
      $("#login-open").classList.toggle("hidden", loggedIn);
      $("#logout").classList.toggle("hidden", !loggedIn);
      $("#user-label").textContent = loggedIn ? currentUser.name : "";
      $("#dashboard-nav").classList.toggle("hidden", !loggedIn || currentUser.role !==
"user");
```

```
    $("#admin-nav").classList.toggle("hidden", !loggedIn || currentUser.role !== "admin");
  }
  document.querySelectorAll("nav [data-page]").forEach(button => button.addEventListener("click", () => go(button.dataset.page)));
  document.querySelectorAll("[data-go]").forEach(button => button.addEventListener("click", () => go(button.dataset.go)));
  $("#book-search").addEventListener("input", renderBooks);
  $("#material-year").addEventListener("change", renderMaterials);
  $("#material-semester").addEventListener("change", renderMaterials);
  $("#login-open").addEventListener("click", () => $("#auth-modal").classList.remove("hidden"));
  $("#modal-close").addEventListener("click", () => $("#auth-modal").classList.add("hidden"));
  $("#auth-modal").addEventListener("click", event => { if (event.target.id === "auth-modal") $("#auth-modal").classList.add("hidden"); });
  $("#show-login").addEventListener("click", () => { $("#login-form").classList.remove("hidden"); $("#register-form").classList.add("hidden"); $("#auth-title").textContent = "Log in"; });
  $("#show-register").addEventListener("click", () => { $("#register-form").classList.remove("hidden"); $("#login-form").classList.add("hidden"); $("#auth-title").textContent = "Create account"; });
  $("#login-form").addEventListener("submit", event => {
    event.preventDefault();
    const data = new FormData(event.currentTarget), email = String(data.get("email")).trim().toLowerCase();
    const user = db.users.find(item => item.email === email && item.password === data.get("password"));
    if (!user) return notice("Email or password is incorrect.", true);
    currentUser = user; updateAuth(); $("#auth-modal").classList.add("hidden"); event.currentTarget.reset();
    go(user.role === "admin" ? "admin" : "dashboard"); notice(`Signed in as ${user.name}.`);
  });
  $("#register-form").addEventListener("submit", event => {
    event.preventDefault();
    const data = new FormData(event.currentTarget), email = String(data.get("email")).trim().toLowerCase();
    if (db.users.some(user => user.email === email)) return notice("An account with this email already exists.", true);
    const user = { name: String(data.get("name")).trim(), email, password: String(data.get("password")), role: "user" };
    db.users.push(user); save(); currentUser = user; updateAuth(); $("#auth-modal").classList.add("hidden"); event.currentTarget.reset();
    go("dashboard"); notice("Your account is ready.");
  });
```

```
    $("#logout").addEventListener("click", () => { currentUser = null; updateAuth();
go("home"); notice("You have logged out."); });
    $("#book-form").addEventListener("submit", event => {
      event.preventDefault(); const data = new FormData(event.currentTarget);
      const id = Number(data.get("bookId"));
      const book = db.books.find(item => item.id === id);
      if (book) {
        const borrowed = book.copies - available(book);
        if (Number(data.get("copies")) < borrowed) return notice(`Copies cannot be
lower than the ${borrowed} currently issued.`, true);
        Object.assign(book, { title: String(data.get("title")).trim(), author:
String(data.get("author")).trim(), subject: String(data.get("subject")).trim(),
copies: Number(data.get("copies")) });
      } else {
        db.books.push({ id: Date.now(), title: String(data.get("title")).trim(),
author: String(data.get("author")).trim(), subject:
String(data.get("subject")).trim(), copies: Number(data.get("copies")) });
      }
      save(); event.currentTarget.reset(); event.currentTarget.elements.copies.min =
1; event.currentTarget.querySelector('[type="submit"]').textContent = "Add book";
$("#cancel-book-edit").classList.add("hidden"); renderAdmin(); renderBooks();
notice(book ? "Book updated." : "Book added to inventory.");
    });
    $("#cancel-book-edit").addEventListener("click", () => {
      const form = $("#book-form"); form.reset(); form.elements.copies.min = 1;
form.querySelector('[type="submit"]').textContent = "Add book";
$("#cancel-book-edit").classList.add("hidden");
    });
    $("#issue-form").addEventListener("submit", event => {
      event.preventDefault(); const data = new FormData(event.currentTarget), bookId =
Number(data.get("book"));
      const book = db.books.find(item => item.id === bookId);
      if (!book || available(book) < 1) return notice("No copies of this book are
currently available.", true);
      db.loans.push({ id: Date.now(), email: String(data.get("user")), bookId, issue:
new Date().toISOString().slice(0,10), due: String(data.get("due")), returned: false
});
      save(); renderAdmin(); notice("Book issued successfully.");
    });
    $("#material-form").addEventListener("submit", event => {
      event.preventDefault();
      const data = new FormData(event.currentTarget), file = data.get("file");
      const id = Number(data.get("materialId"));
      const material = db.materials.find(item => item.id === id);
      const finish = (fileData, filename) => {
        const values = { title: String(data.get("title")).trim(), year:
String(data.get("year")), semester: String(data.get("semester")) };
```

```
        if (material) Object.assign(material, values, fileData === null ? {} : { data:
fileData, filename });
        else db.materials.push({ id: Date.now(), ...values, filename, data: fileData
});
        save(); event.currentTarget.reset();
event.currentTarget.querySelector('[type="submit"]').textContent = "Add material";
$("#cancel-material-edit").classList.add("hidden"); renderAdmin(); renderMaterials();
notice(material ? "Study material updated." : "Study material added.");
      };
      if (file instanceof File && file.size) {
        const reader = new FileReader();
        reader.onerror = () => notice("Could not read the selected material file.",
true);
        reader.onload = () => finish(reader.result, file.name);
        reader.readAsDataURL(file);
      } else if (material) finish(null, material.filename);
      else notice("Choose a file to add the study material.", true);
    });
    $("#cancel-material-edit").addEventListener("click", () => {
      const form = $("#material-form"); form.reset();
form.querySelector('[type="submit"]').textContent = "Add material"; $("#cancel-
material-edit").classList.add("hidden");
    });
    $("#contact-form").addEventListener("submit", event => {
      event.preventDefault(); event.currentTarget.reset();
      notice("Thanks for your message. In this demo, connect the form to an email
service to deliver it.");
    });
    $("#featured-books").innerHTML = db.books.slice(0,3).map(bookCard).join("");
    renderBooks(); renderMaterials(); updateAuth();
  </script>
</body>
</html>
```

OUTPUT:


New Tab
Cornerstone Library
Harborlight Hotel | Stay by the
Ask Gemini
File
/Users/Tuco/Desktop/hotel-booking.html
Finish update
All Bookmarks
HARBORLIGHT HOTEL
Home
Rooms
About
Local guide
Help & contact
Privacy
Log in
STAY A WHILE
Our rooms
Comfortable spaces, simple rates, no surprises.
Classic Room
A welcoming room with everything you need for a comfortable stay.
Up to 2 guests
$119 / night
Book room
Deluxe Room
More room to unwind, with a sitting area and garden views.
Up to 3 guests
$169 / night
Book room
Family Suite
Room for the whole crew, with separate sleeping and lounge areas.
Up to 4 guests
$229 / night
Book room
Ocean Suite
A special-occasion suite with a private balcony and sea views.
Up to 4 guests
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HARBORLIGHT HOTEL
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Log in
OUR STORY
About Harborlight
Harborlight is a small, independent coastal hotel built around a simple idea: a good trip starts with feeling at home. Our friendly team is here to help you settle in, explore, and enjoy your time by the water.
What makes a Harborlight stay?
Complimentary Wi-Fi · Breakfast options · 24-hour front desk · Smoke-free rooms
Harborlight Hotel · A friendly place to land · Questions about a stay? Call +1 (555) 010-2040 or email hello@harborlight.example. Our front desk is open 24 hours.

New Tab | Cornerstone Library | Harborlight Hotel | Stay by the | Ask Gemini

File /Users/Tuco/Desktop/hotel-booking.html | Finish update | All Bookmarks

HARBORLIGHT HOTEL

Home Rooms About Local guide Help & contact Privacy Log in

AROUND THE NEIGHBORHOOD

Local travel guide

The historic harbor is a 10-minute walk from the hotel. Visit the weekend farmers market, follow the coastal path to Lighthouse Point, or ask our front desk for local bus schedules. Parking is available on-site, subject to availability.

Harborlight Hotel · A friendly place to land · Questions about a stay? Call +1 (555) 010-2040 or email hello@harborlight.example. Our front desk is open 24 hours.



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WE'RE HERE TO HELP

Help & contact

Contact us

Questions about a stay? Call +1 (555) 010-2040 or email hello@harborlight.example. Our front desk is open 24 hours.

Your name

Email

Message

Send message

Frequently asked questions

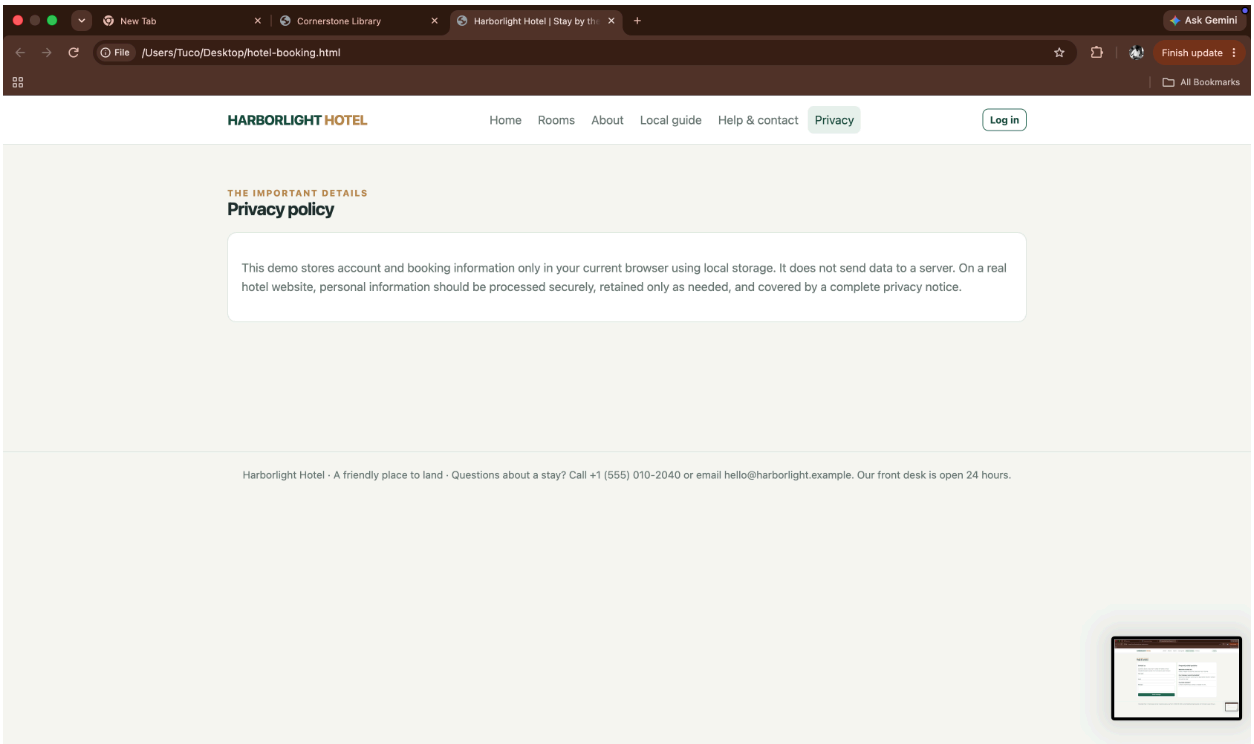
What time is check-in?
Check-in begins at 3:00 PM; check-out is by 11:00 AM.

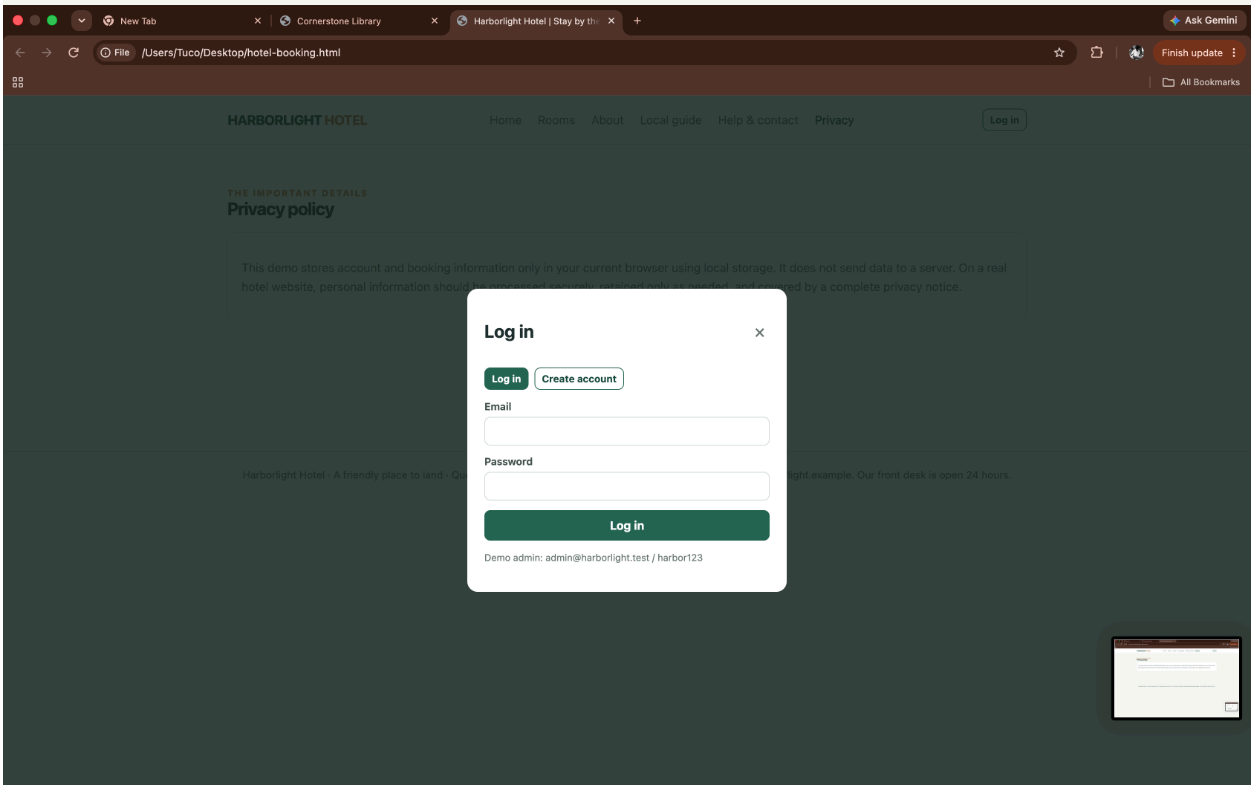
Can I change or cancel my booking?
Guests can change or cancel up to 3 days before check-in. Contact the hotel for help.

Is parking available?
Limited complimentary parking is available on-site.

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Rooms
About
Local guide
Help & contact
Privacy
Log in
THE IMPORTANT DETAILS
Privacy policy
This demo stores account and booking information only in your current browser using local storage. It does not send data to a server. On a real hotel website, personal information should be processed securely, retained only as needed, and covered by a complete privacy notice.
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Log in
THE IMPORTANT DETAILS
Privacy policy
This demo stores account and booking information only in your current browser using local storage. It does not send data to a server. On a real
Log in
Log in
Create account
Email
Password
Log in
Demo admin: admin@harborlight.test / harbor123

RESULT : The Online Hotel Room Booking Website was developed to simplify hotel reservation and management activities. Users can register, log in, manage their profiles, check room availability, book rooms, view booking history, and modify or cancel reservations according to the three-day policy. Administrators can update room quantities and prices, modify or cancel bookings, and manage static website pages such as room information, About Us, Contact Us, customer service FAQs, local travel guides, and privacy policies. The website provides a user-friendly and responsive interface, making hotel booking and administration more convenient and efficient.